File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch

SIDE PROXI SENSOR

REAR-IR-R
REAR-IR-L
HALL-TOWER
HALL-BIN
IR-R-RX
VL-L-INT
SDA
SCL

CLIFF

CLIFF-RM-Q+
CLIFF-LM-Q+
RR-Q+
RF-Q+
LR-Q+
LF-Q+
BUMP-R-Q+
BUMP-L-Q+
BUMPER_LED_EN
CLIFF_LED_EN

File: CLIFF.kicad_sch

MAIN BRUSH MOTOR

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADC

File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs

WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-AD
WHEEL-M-RIGHT-DROPD
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROPD
WHEEL-M-LEFT-ENCODE-AD
WHEEL-M-R-F-ADCD
WHEEL-M-L-F-ADCD

File: WHEEL-MOTORs .kicad_sch

MAIN FAN

MAIN-FAN-S-SENSED
MAIN-FAN-S-CTRLD

File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs

SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADCD

File: SIDE-BRUSH-MOTORs .kicad_sch

MCU-STM32

DC_IN_SENSE
CLIFF-LM-Q+
CLIFF-RM-Q+
HALL-BIN
HALL-TOWER
REAR-IR-L
REAR-IR-R
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADC
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-RIGHT-DROP
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROP
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
MAIN-FAN-S-SENSE
MAIN-FAN-S-CTRL
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADC
CARPET-SENSOR-ECHO
CARPET-SENSOR-LO
WDI
LIDAR_RX
LIDAR_TX
LIDAR-M-CTRL
UART3_RX2
UART3_RX1
VL-L-INT
VL-L-LPn
VL-I2C-RESET
USB_CC1
USB_CC2
USB_VBUS_SENSE
VM-VBAT-EN
IMU-INT1
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
POWER-BTN
PWR-SW-G
WD_OK
I2C4_SDA
I2C4_SCL

File: STM32G473.kicad_sch

IMU

IMU-SDAD
IMU-SCLD
IMU-INT#1D

File: IMU-ICM-4267-P .kicad_sch

BUTTON-LEDs

LED-HOMED
LED-PWRD
CLEAN-BTND
HOME-BTND
POWER-BTND

File: BUTTON-LEDs .kicad_sch

POWER CONTROL

STM-PWR-CTRL
PWR-BTN-SENSE
5V_EN
3V_EN
WD_OK
PWR-SW-G
POWER-BTN
VM-VBAT-EN

File: SYSTEM-POWER.kicad_sch

BATTERY CHARGER

BATTERY-CHARGER
DC_IN_SENSE
USB_CC1
USB_CC2
USB_VBUS_SENSE
PWR-SW-G

File: Battery-Charger.kicad_sch

CARPET-SENSOR

CARPET-SENSOR
ECHO_ADC
LO

File: Carpet-sensor.kicad_sch

WD

WATCHDOG
WD_OK
WDI

File: WATCHDOG.kicad_sch

LIDAR

LIDAR-RXD
LIDAR-TXD
LIDAR-MOTOR-CTRL
LIDAR-PWR
WD_OKD
VM-VBAT-EN

File: LIDAR.kicad_sch

FRONT SENSORS

FR-F-RX2
FR-F-RX1
VL-L-INT
VL-L-LPn
VL-R-INT
VL-I2C-RESET
VL-I2C-SCL
VL-I2C-SDA

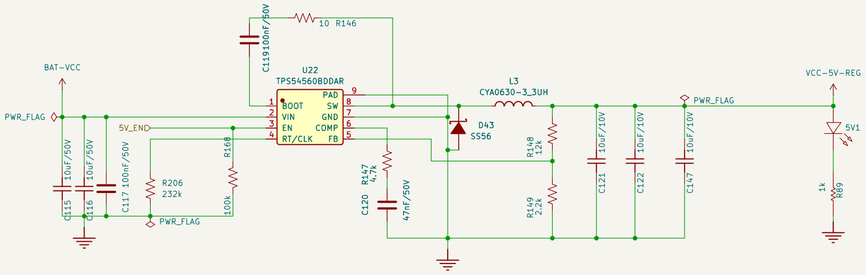
File: Front-Sensors.kicad_sch

Sheet: /
File: Main.kicad_sch

Title:

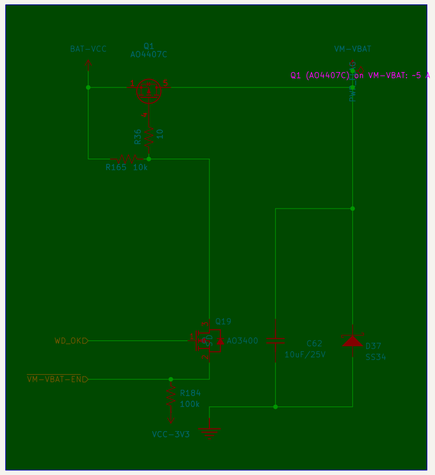
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 1/16

POWER MANAGMENT



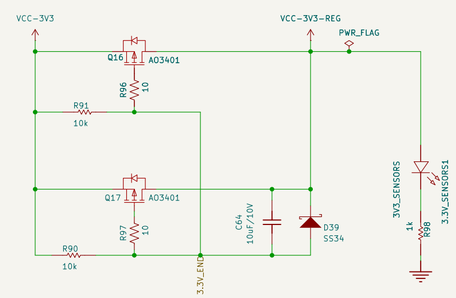
PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.

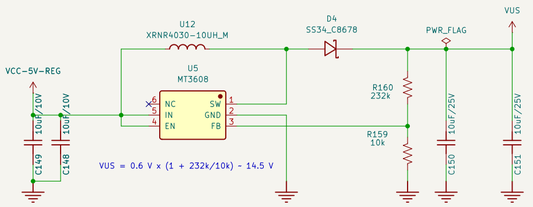




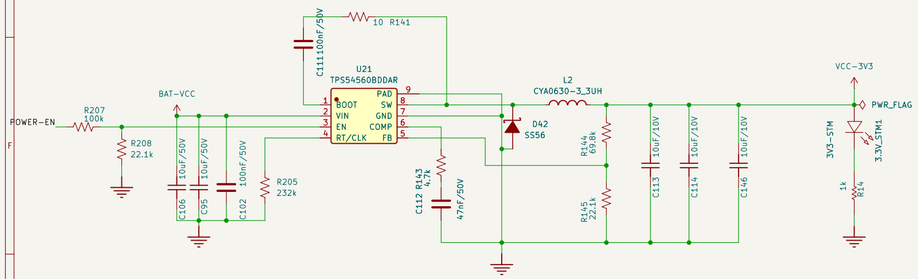
Old VM-5V switch removed: the LiDAR (VM-5V-LIDAR) and the water pump (VM-5V-WATER-PUMP) now have their own WD_OK-gated switches on their sheets.
No soft ramp.

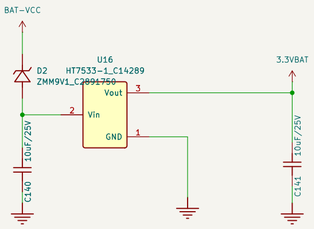
VM-VBAT (motors, fan) is on only while WD_OK is high (RC watchdog fed) AND $\overline{\text{VM-VBAT-EN}}$ is low.
Q19 source-switched AND: GPIO low = on. R184 keeps it off in reset/flash. AO4407C: Vgs +/-25 V, no gate divider needed.

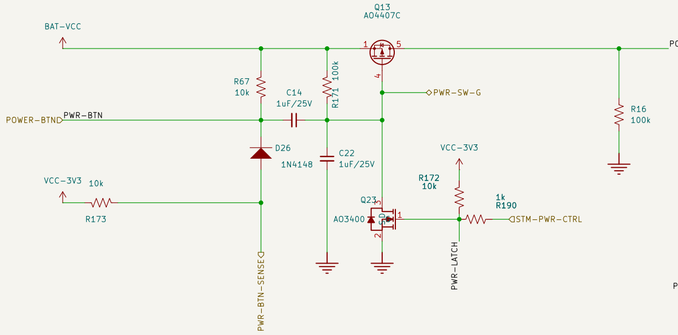




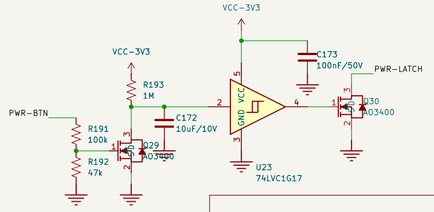
TPS54560B (U21, U22): non-synchronous,
SS56 catch diode: fsw 418 kHz (RT 232k);
COMP 4.7k + 47nF (all-ceramic Cout, no pole cap).
EN abs max 8.4 V: R207/R208 on U21.







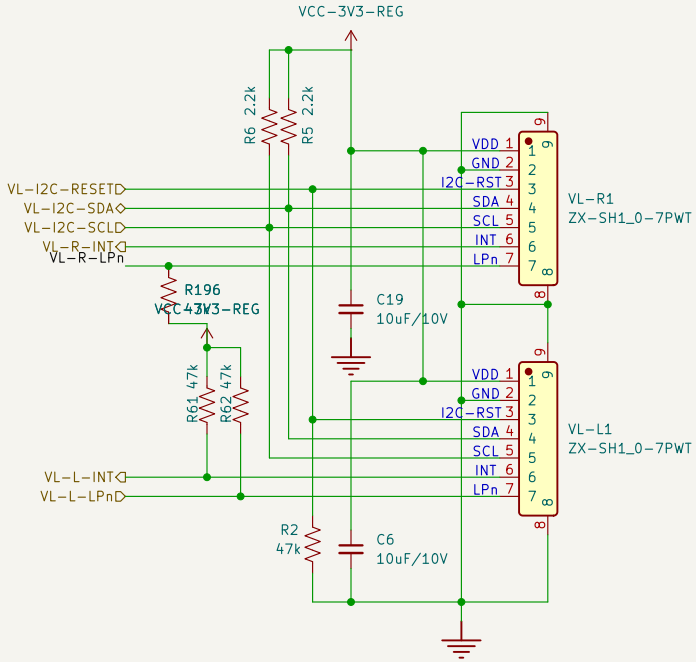
STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board



Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR-LATCH low). Firmware turns off at 3 s.
While charging, the rail-hold (Q31 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR-SW-G low).

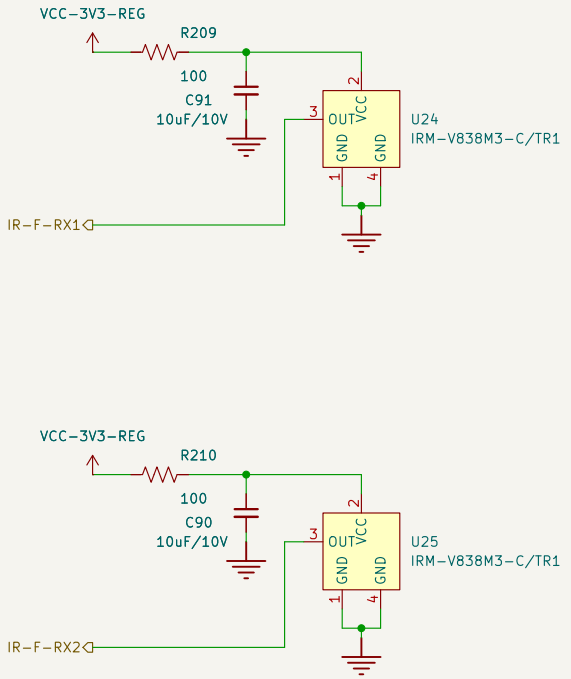
Sheet: /POWER/		
File: SYSTEM-POWER.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/16

2x VL53L7 ToF BOARDS



2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window.

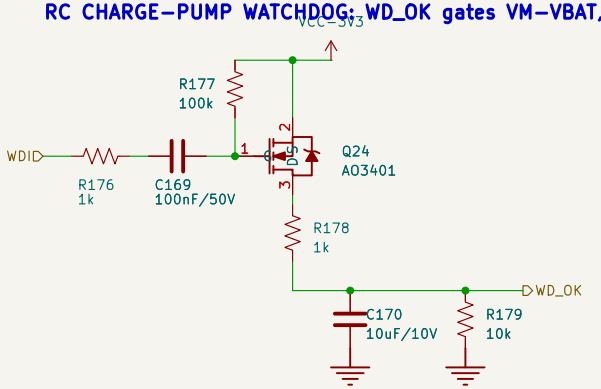


Sheet: /FRONT SENSORS/
File: Front-Sensors.kicad_sch

Title:

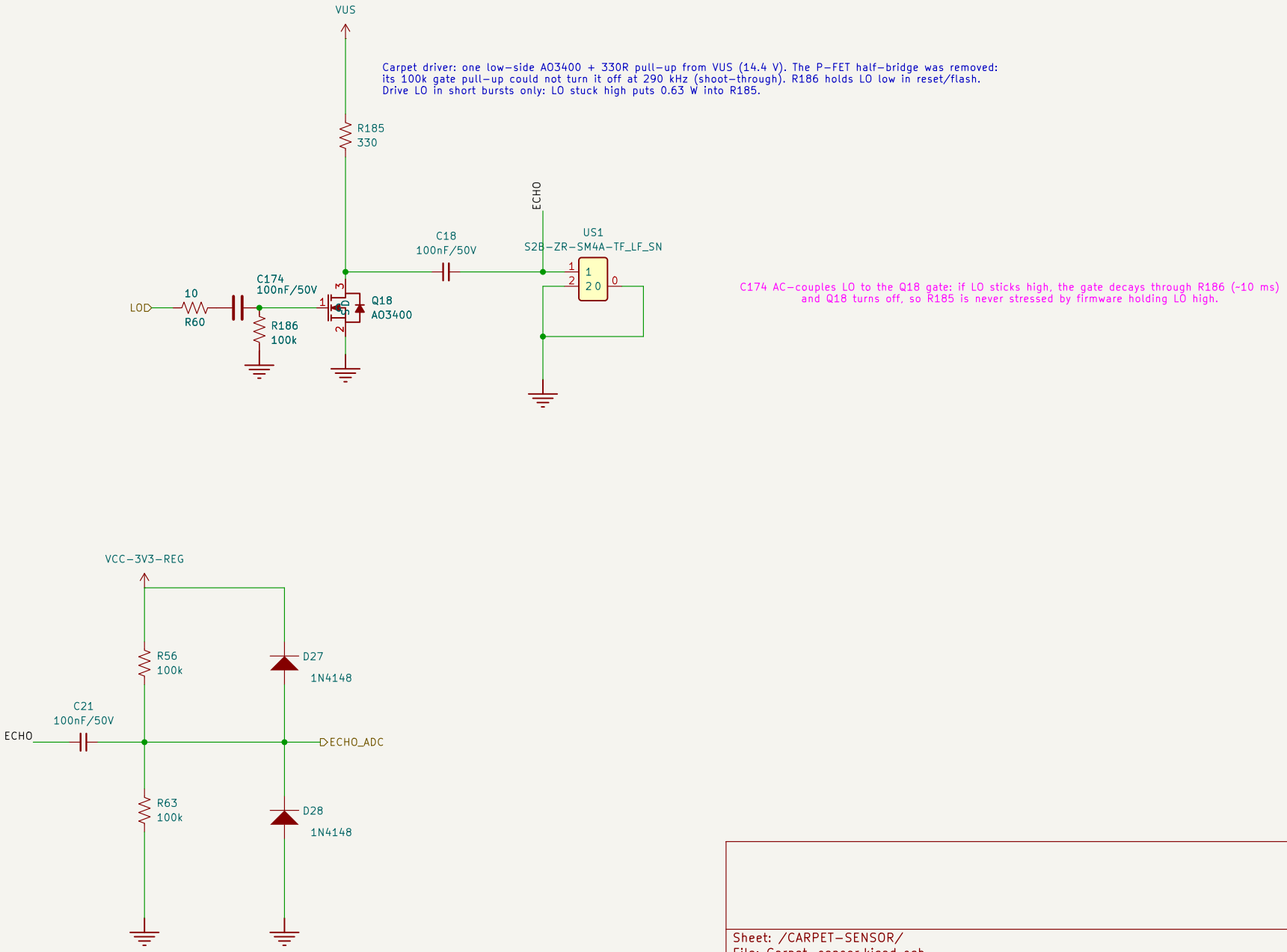
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 4/16

RC CHARGE-PUMP WATCHDOG: WD_OK gates VM-VBAT, VM-5V-LIDAR and VM-5V-WATER-PUMP



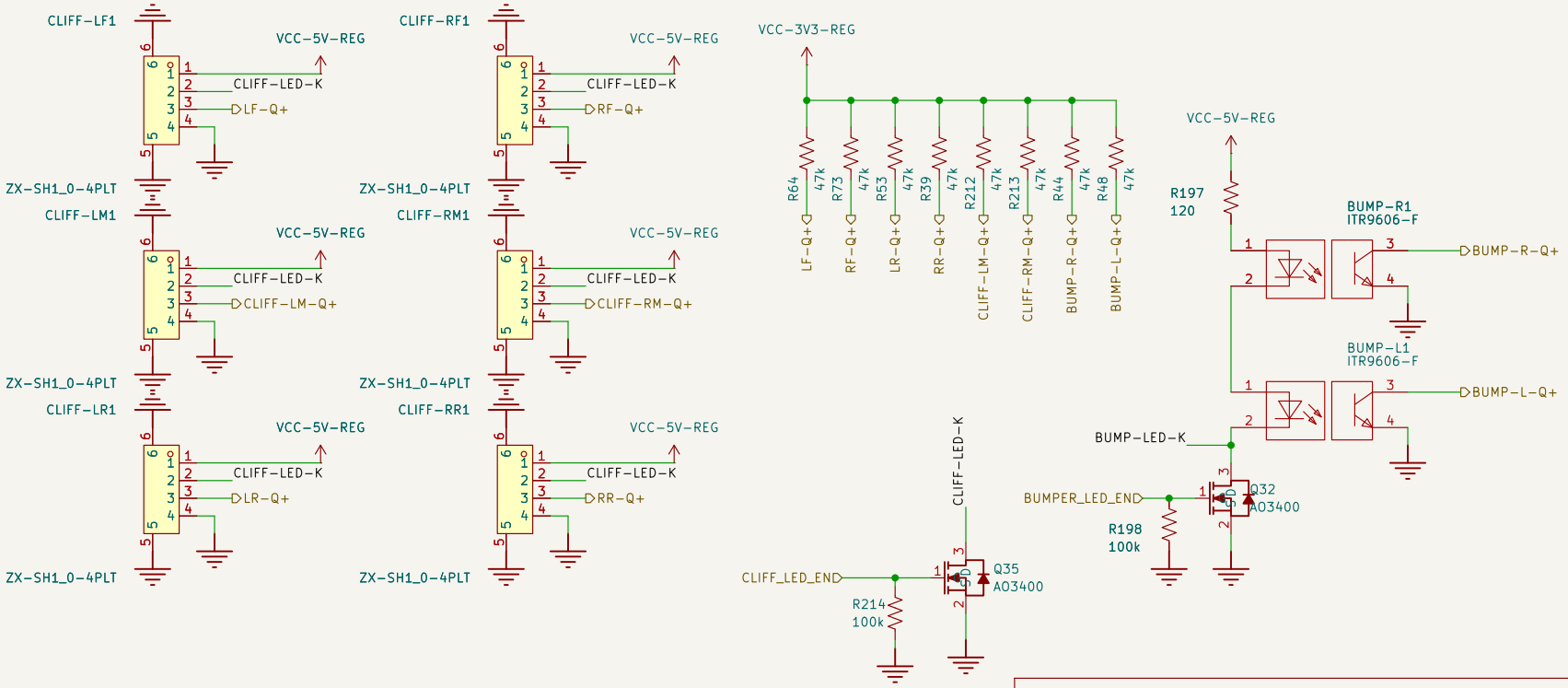
Firmware toggles WDI in software (50–1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/16



Sheet: /CARPET-SENSOR/	
File: Carpet-sensor.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 6/16

ANTI FALL SENSORs



Cliff sensors: 6 boards (kicad/cliff-sensor), SH 4-pin vertical: 1 VCC-5V-REG (LED, 150R on the board), 2 CLIFF-LED-K (Q35 sinks all 6 LEDs), 3 phototransistor collector (47k pull-up), 4 GND.
Modulate CLIFF_LED_EN and read each channel LED-on minus LED-off to cancel ambient light.
ADC: LF PA3, RF PA1, LR PA2, RR PA0, LM PD9, RM PE8 (LM/RM via global labels).

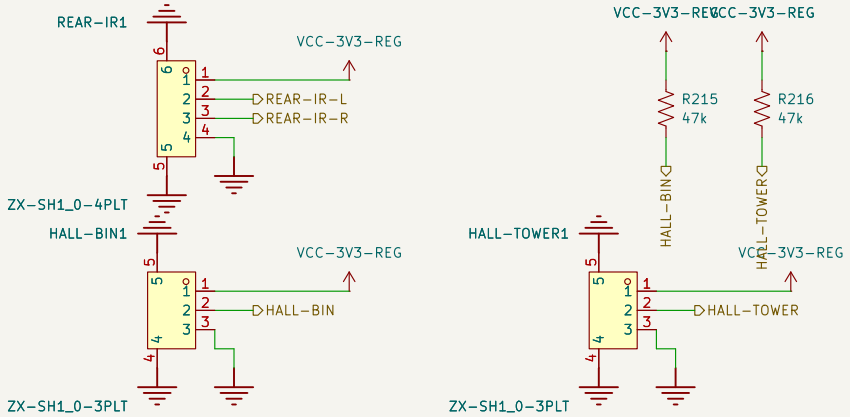
Bumpers: 2x ITR9606 slotted photo-interrupters on the main board (~11 mm tall, set by the bumper mechanism). Emitters in series from VCC-5V-REG, switched by BUMPER_LED_EN (high = on, 100k pull-down). BUMP-x-Q+ reads low when light passes the slot (R44/R48 47k pull-ups).

Sheet: /CLIFF/	
File: CLIFF.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 19/16



SIDE PROXI IR SENSORs

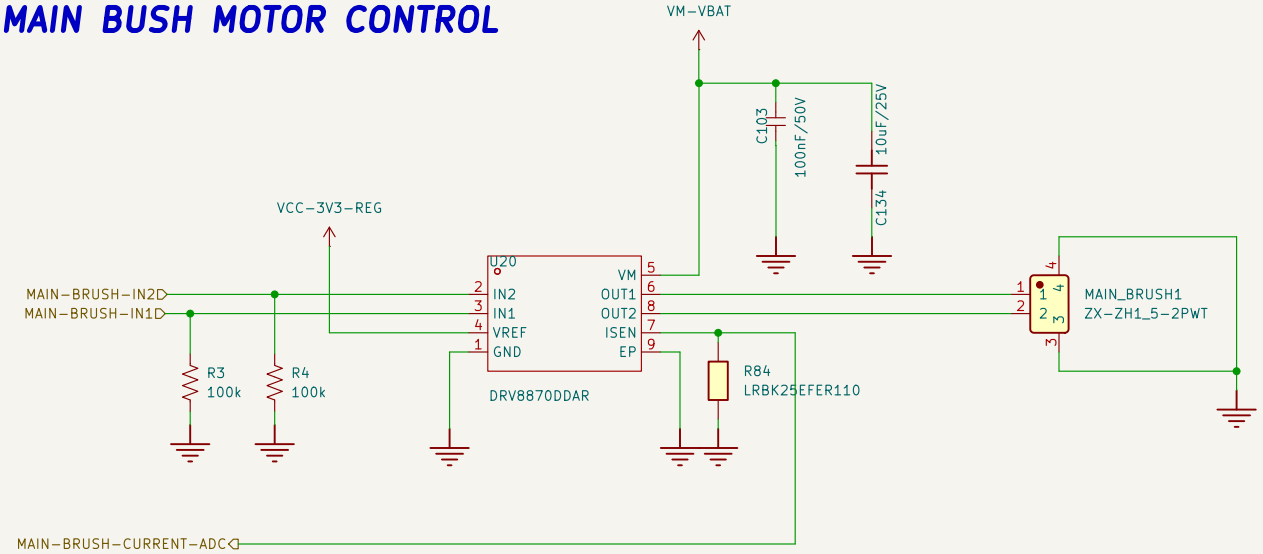
REAR DOCKING IR + HALL SENSORS



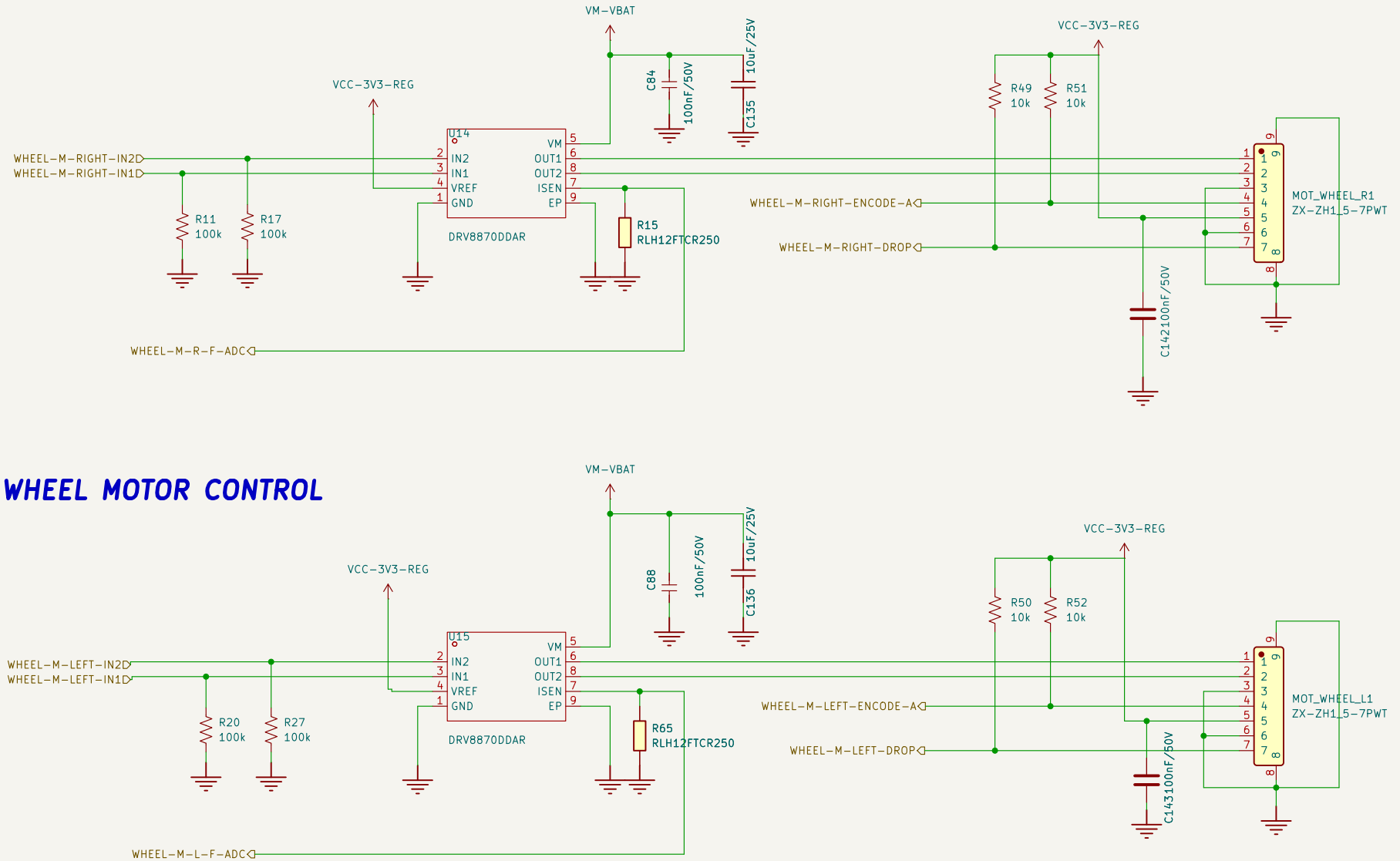
REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

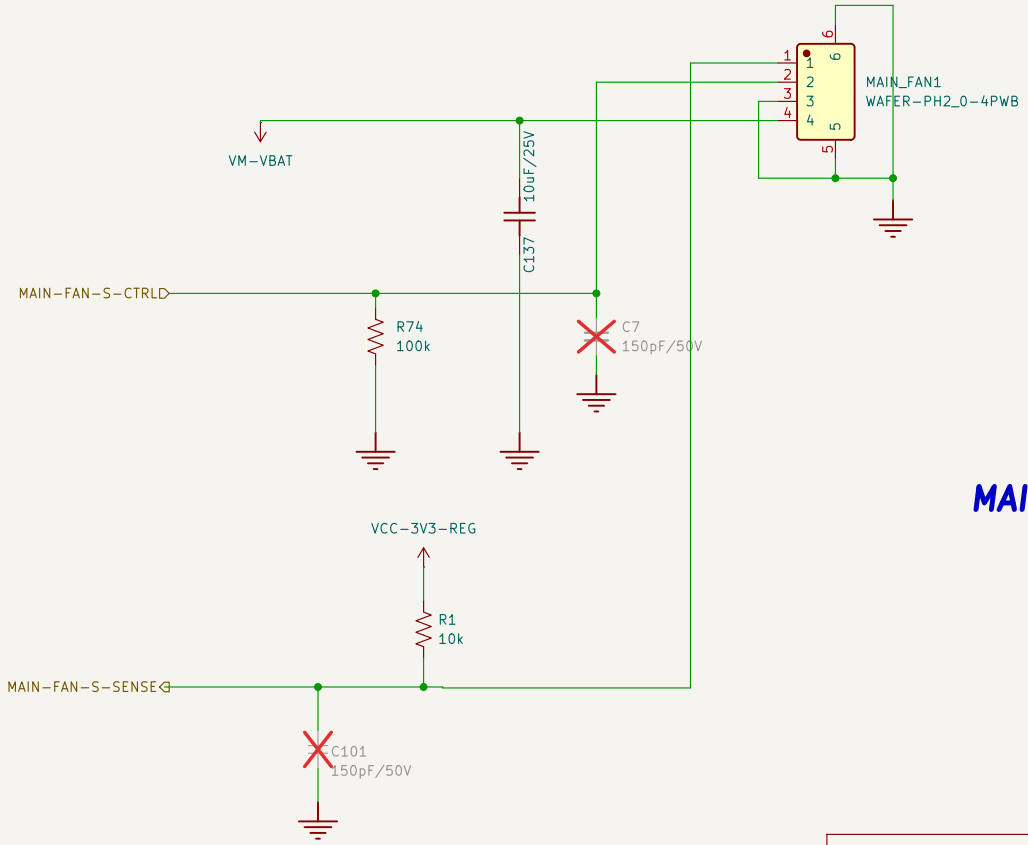
Sheet: /SIDE-PROXI/	
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 20/16

MAIN BUSH MOTOR CONTROL



Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/16

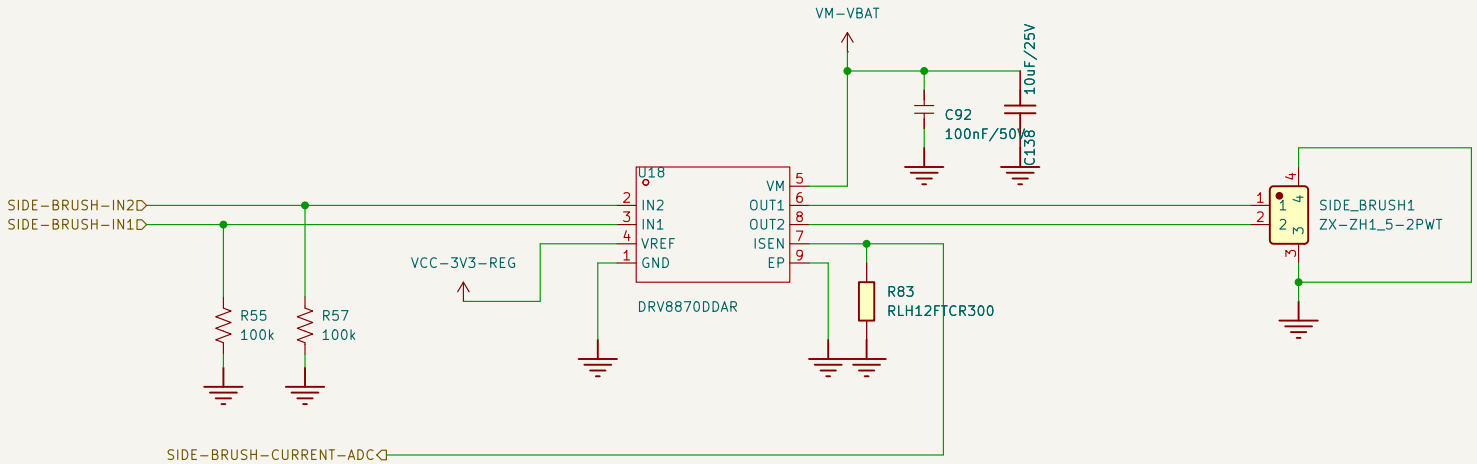

WHEEL MOTOR CONTROL
VM-VBAT
VCC-3V3-REG
U14
IN2
IN1
VREF
GND
VM
OUT1
OUT2
ISEN
EP
DRV8870DDAR
C84
100nF/50V
10uF/25V
C135
R11
100k
R17
100k
R15
RLH12FTCR250
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-R-F-ADC
R49
10k
R51
10k
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-RIGHT-DROP
C142100nF/50V
MOT_WHEEL_R1
ZX-ZH1.5-7PWT
U15
C88
C136
R20
R27
R65
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-L-F-ADC
R50
10k
R52
10k
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-LEFT-DROP
C143100nF/50V
MOT_WHEEL_L1
Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch
Title:
Size: USLetter
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 23/16



MAIN FAN CONTROL

Sheet: /MAIN-FAN/		
File: MAIN-FAN .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/16

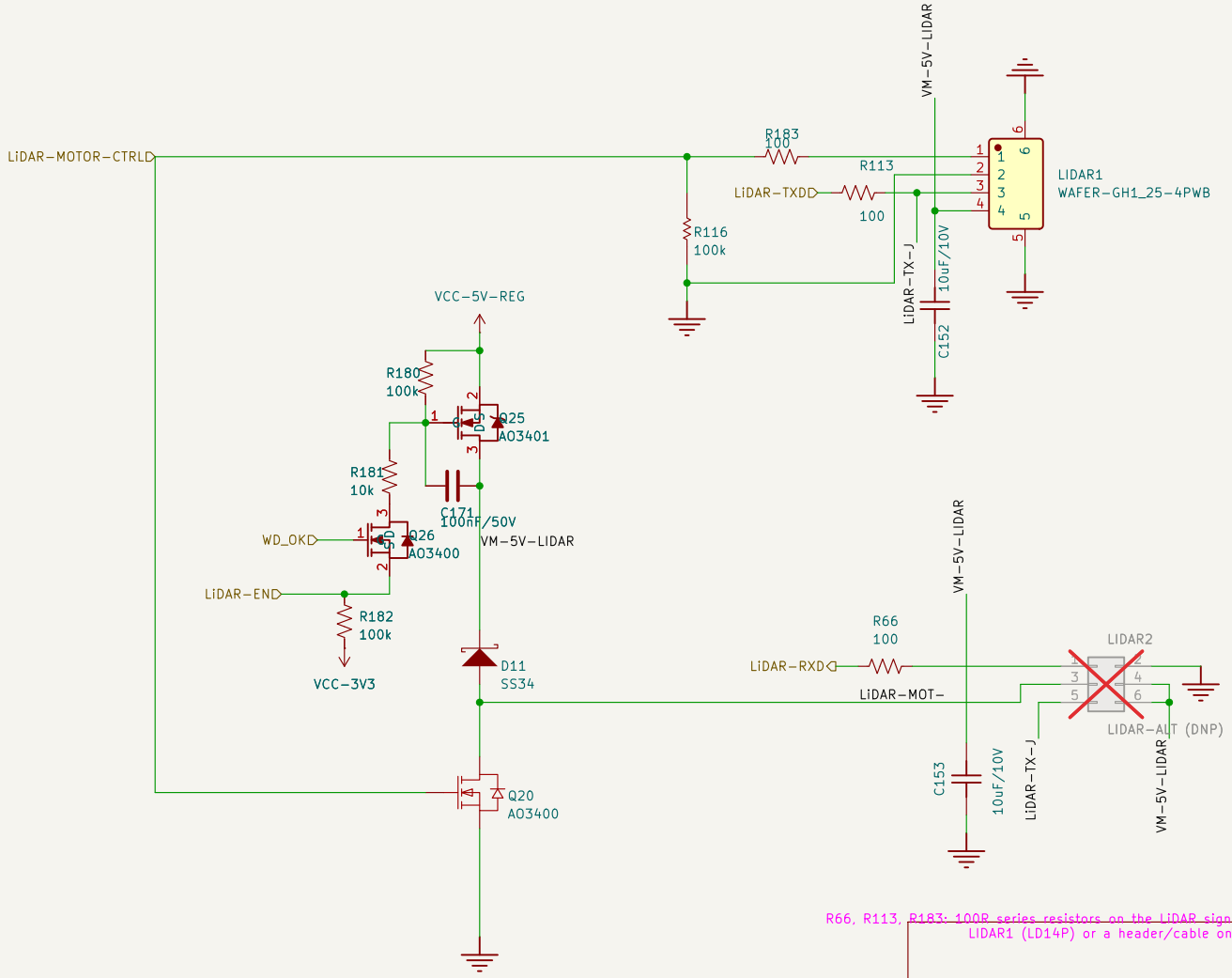
SIDE-BRUSH MOTOR CONTROL



Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 25/16

VM-5V-LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR-EN is low (Q26 source-switched AND). R181/C171: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 2x3 0.1in through-hole pads (not fitted) for other LiDARs. Pin 6 = VM-5V-LIDAR for LiDARs
that need a separate motor supply (motor between pins 6 and 3). Pin 5 = the LiDAR TX, shared with LIDAR1 pin 3.

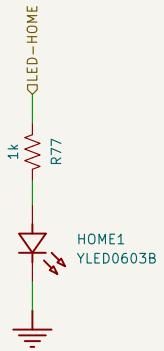
LiDAR

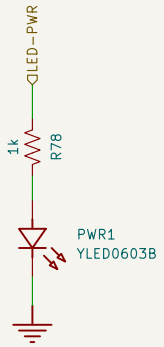


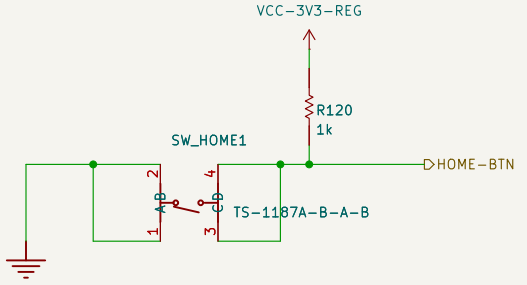
R66, R113, R183: 100R series resistors on the LiDAR signals. Only one LiDAR is fitted:
LIDAR1 (LD14P) or a header/cable on LIDAR2.

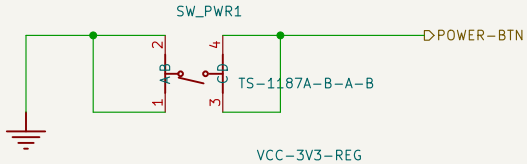
Sheet: /LiDAR/	
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Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 26/16

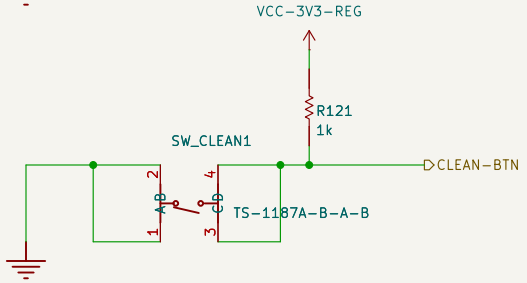
LEDs AND BUTTONs











Sheet: /BUTTON-LED/	
File: BUTTON-LEDs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 28/16

Phone fork: PB6/PB4 = UCPD1 CC1/CC2 (USB-C), PB5 = 5V_EN, PA7 = USB_VBUS_SENSE (ADC2_IN4).
Front ToF moved from the Pi: I2C on I2C4 (shared with the side boards), PC0 = VL-I2C-RESET,
PD12 = VL-L-INT, PC8 = VL-R-INT, PB13 = VL-L-LPn, PE8 = VL-R-LPn. Use internal pull-ups on INT.

U13A
STM32G473VCT6

RR-Q+ 20 PA0
RF-Q+ 21 PA1
LR-Q+ 22 PA2
LF-Q+ 25 PA3
BUMP-L-Q+ 26 PA4
BUMP-R-Q+ 27 PA5
MAIN-BRUSH-CURRENT-ADC 28 PA6
USB_VBUS_SENSE 29 PA7
SPARE-PA8 69 PA8
SPARE-PA9 70 PA9
SPARE-PA10 71 PA10
DM 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15

WHEEL-M-RIGHT-DROP 32 PB0
WHEEL-M-LEFT-DROP 33 PB1
NTC 34 PB2
STM32-SWO 90 PB3
USB_CC2 91 PB4
5V_EN 92 PB5
USB_CC1 93 PB6
WHEEL-M-RIGHT-IN1 94 PB7
BOOT0 95 PB8-BOOT0
WHEEL-M-RIGHT-IN2 96 PB9
3.3V_EN R195 1K 47 PB10
UART3_RX2 10 R80 50 PB11
51 PB12
VL-L-LPn 52 PB13
BAT_ID 53 PB14
DC_IN_SENSED 54 PB15

PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LiDAR-M-CTRL
PC3 18 BUMPER_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9

PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-S-CTRL
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM-Q+
PD10 57 WHEEL-M-R-F-ADC
PD11 58 WHEEL-M-L-F-ADC
PD12 59 VL-L-INT
PD13 60 SIDE-BRUSH-CURRENT-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2

VCC-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
LDR
R103 10k
R22 10k
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V

U13B
STM32G473VCT6

WHEEL-M-LEFT-IN1 97 PE0
WHEEL-M-LEFT-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIND 3 PE4
HALL-TOWER 4 PE5
SPARE-PE6 5 PE6
38 PE7
PWR-BTN-SENSED 39 PE8
CLIFF-RM-Q+ 40 PE9
MAIN-FAN-S-SENSED 41 PE10
VM-VBAT-EN 42 PE11
CARPET-SENSOR-LO 43 PE12
CARPET-SENSOR-ECHO 44 PE13
WHEEL-M-LEFT-ENCODE-AD 45 PE14
WHEEL-M-RIGHT-ENCODE-AD 46 PE15
UART3_RX1

VCC-3V3
R21 10k
X1 ABM3-8_000MHZ-D2Y-T
12 PF0-OSC_IN
13 PF1-OSC_OUT
19 PF2 LED-PWR
10 PF9 SIDE-BRUSH-IN1
11 PF10 SIDE-BRUSH-IN2
14 PG10-NRST RST
C37 100nF/50V
C34 22pF/50V
C35 22pF/50V

U13C
STM32G473VCT6

VCC-3V3
24 49 64 75 100 VDD
3.3VBAT
37 VDDA
6 VBAT
23 48 63 74 99 VSS
35 VSSA
36 VREF+
VCC-3V3

STM32_DEBUG1
STM32-RX
STM32-TX
PZ254V-11-05P
Compatible with TTL-234X-3V3 smart cable

JTAG1
PZ1_27-2X7P-TP
VCC-3V3
R40 100k
STM32-TX 14 13 STM32-RX
RST 12 11
10 9
STM32-SWO 8 7
STM32-SWCLK 6 5
STM32-SWDIO 4 3
2 1

Compatible with STDC14

Pin 11 (GNDDetect) can detect
JTAG cable presence using a 100k
pull-up plus a spare GPIO

The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors

J1 spare I/O: 0.1in 2x7 header (fitted).
1 VCC-3V3-REG (sensor rail), 2 VCC-5V-REG, 13/14 GND.
3/4 not connected (I2C4 kept off the header: 1 MHz bus shared with the ToF, side sensor and IMU).
5/6 UART PC10 TX / PC11 RX (UART4 or USART3). 7 PA10, 8 PE6, 9 PA8, 10 PA9 (10k pull-up), 11 PD4, 12 PE2.
IMU AD0/INT2/FSYNC are test pads on the IMU sheet. SWO (PB3) on JTAG1 is a spare when not tracing.

R204: PA9 = UCPD DBCC1. High during reset
turns on the dead-battery Rd on CC1 (PB6).

VCC-3V3-REG
R204 5k1
VCC-5V-REG
J1
SPARE-PC10-TX SPARE-PC11-RX
SPARE-PA10 SPARE-PE6
SPARE-PA8 SPARE-PA9
SPARE-PD4 SPARE-PE2
SPARE-IO

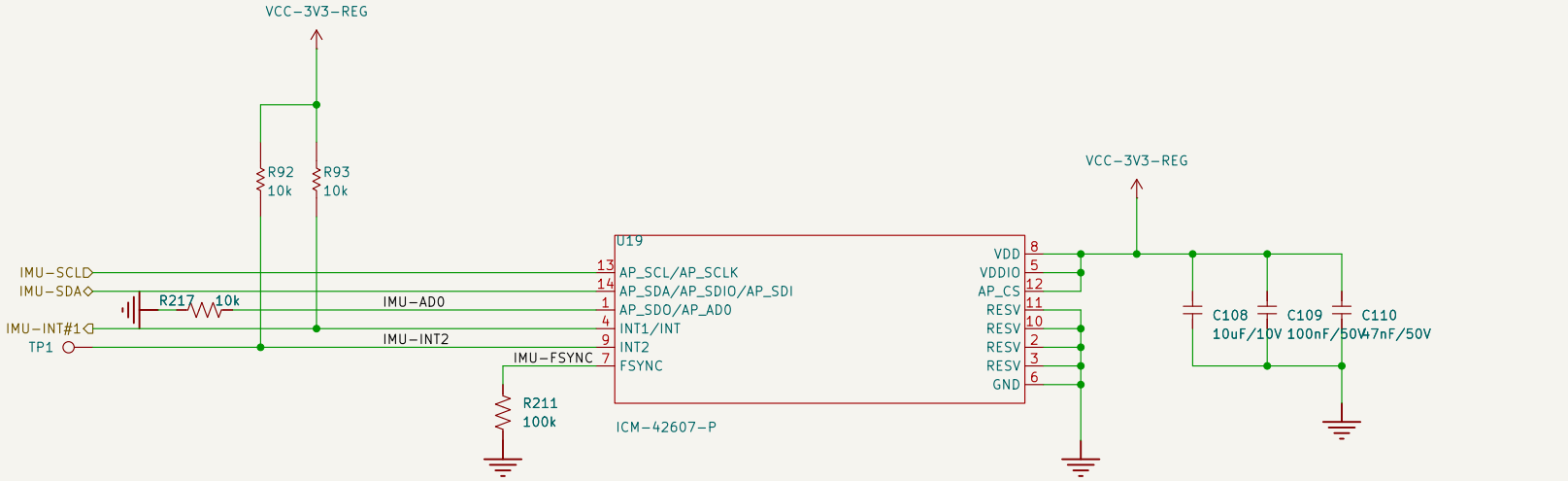
VCC-3V3
PWR_FLAG
3.3VBAT
VCC-3V3
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C36 100nF/50V
C32 100nF/50V
C33 10uF/10V

Sheet: /MCU-STM32/
File: STM32G473.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 30/16

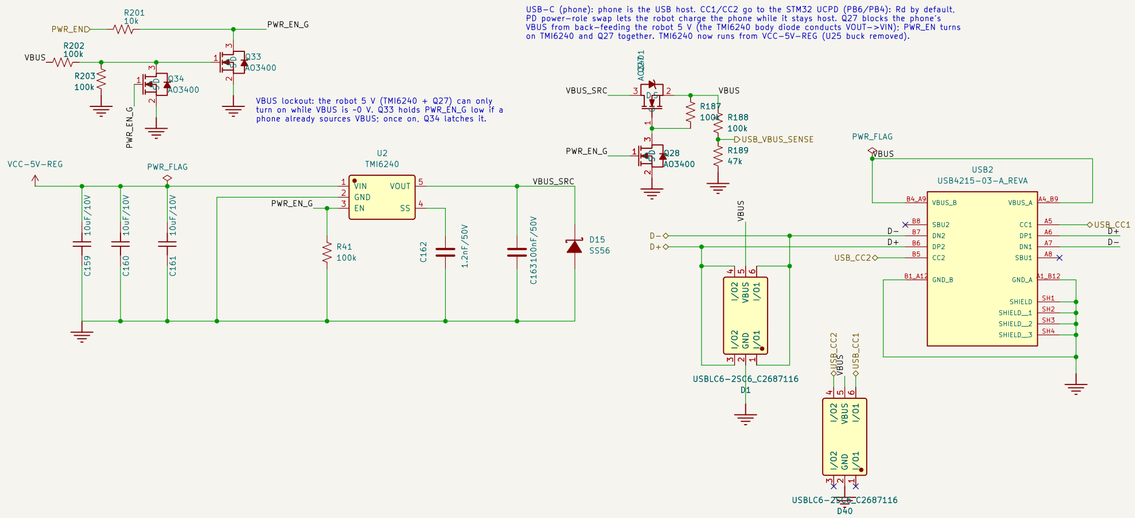
IMU-ICM-42607-P



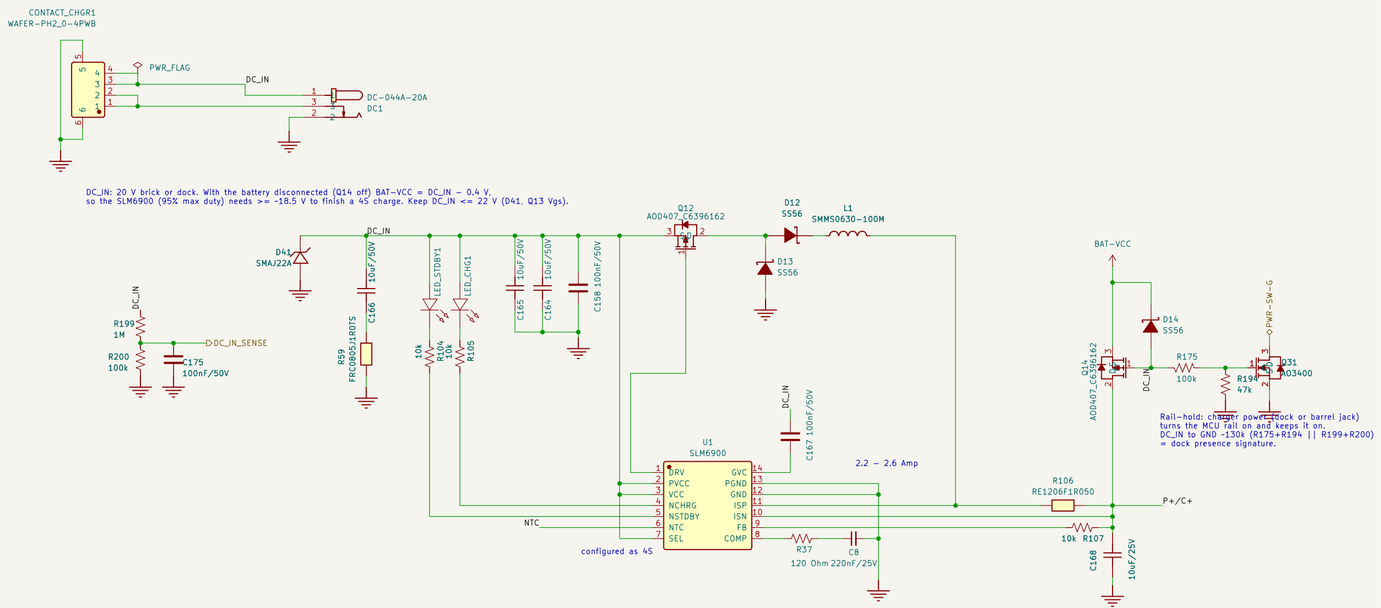
IMU on I2C4 (2.2k pull-ups on the MCU sheet). AP_AD0 pulled low by R217 (I2C address 0x68). TP1 = INT2 (test pad; the IMU is leadless). Probe AD0 on R217 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

Sheet: /IMU/	
File: IMU-ICM-4267-P .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 32/16

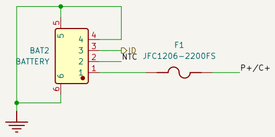
USB-C



BATTERY CHARGER



BATTERY CONNECTOR



Sheet: /BATTERY-CHARGER/	
File: Battery-Charger.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 33/16